

Galois Theory

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Autumn 2025

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1 Motivation

1.1 Solving harder polynomials

Corti's way of teaching Galois theory is a charming combination of categorifying the main theory, while involving lots of computational methods. It is probably wise to try and approach his spartan expectation of your command of algebraic manipulation.

Proposition 1.1.1. Quadratic equations over a field k can be *solved by radicals*.¹

Proof. Have $aX^2 + bX + c \in k[X]$, and execute the following chain of algebra:

$$aX^2 + bX + c = a \left(X - \frac{b}{2a} \right)^2 - \frac{b^2}{2a} + c = 0 \iff X - \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \quad \square$$

Proposition 1.1.2. Cubic equations over a field k can be solved by radicals.

Proof. Have $aX^3 + bX^2 + cX + d \in k[X]$, we first want to get rid of the X^2 by “completing the cube”

$$aX^3 + bX^2 + cX + d = a \left(X - \frac{b}{3a} \right)^3 - \left(\frac{b^2X}{3a} + \frac{b^3}{27a^3} \right) + cX + d$$

So if we divide through by a and substitute $Y = X - b/3a$, we get a **depressed cubic** of the form $Y^3 + pY + q = 0$. I'll then use the substitution $Y = u - p/3u$.²

$$\left(u - \frac{p}{3u} \right)^3 + p \left(u - \frac{p}{3u} \right) + q = u^3 - pu + \frac{p^2}{3u} - \frac{p^3}{27u^3} + pu - \frac{p^2}{3u} + q = u^3 + q - \frac{p^3}{27u^3}$$

This is now a quadratic in u^3 , so we can use the previous method and find 6 values for u , only 3 of these will be distinct, we can then substitute back into Y and then X . $\square$

¹This has a precise definition we *might* talk about later on in this course, but you have an intuitive idea of what it really means.

²I know this to be called “Vieta's substitution”.

Proposition 1.1.3. Quartic equations over a field k can be solved by radicals.

Proof. Let's just let this one be monic $f(X) = X^4 + mX^3 + pX^2 + qX + r \in k[X]$ and do some algebra:

$$\begin{aligned} \left(X^2 + \frac{m}{2}X + \frac{\lambda}{2}\right)^2 &= X^4 + mX^3 + \left(\frac{m^2}{4} + \lambda\right)X^2 + \frac{m\lambda}{2}X + \frac{\lambda^2}{4} \\ \implies \left(X^2 + \frac{m}{2}X + \frac{\lambda}{2}\right)^2 - \left[\left(\frac{m^2}{4} + \lambda - p\right)X^2 + \left(\frac{m\lambda}{2} - q\right)X + \left(\frac{\lambda^2}{4} - r\right)\right] &= f \end{aligned}$$

I want to make the right quadratic a perfect square, so I need the discriminant to be zero:

$$\left(\frac{m\lambda}{2} - q\right)^2 - 4\left(\frac{m^2}{4} + \lambda - p\right)\left(\frac{\lambda^2}{4} - r\right) = 0$$

This is a cubic in λ , so we can use the previous method to solve it. Substituting, probably the nicest, solution for λ back in lets us write our expression for f as the difference of two squares, factoring our quartic as the product of two quadratics which we can solve with the first method.³ $\square$

I have not really proved that the last two methods work, do my λ and u always exists, do the multiple roots all collapse nicely, how did people come up with this? I'm content not knowing any of these details and just using these algorithms, because they do in fact always work.

1.2 Field stuff

In this section, fix a field $\mathbb{Q} \subset K \subset \mathbb{C}$. For some $\alpha \in \mathbb{C}$ we will use the notation:

$$K(\alpha) := \left\{ \frac{P(\alpha)}{Q(\alpha)} \in \mathbb{C} \mid P, Q \in K[X], Q(\alpha) \neq 0 \right\}.$$

$K(\alpha_1, \dots, \alpha_n)$ is defined recursively.

Definition 1.2.1. Such an $\alpha \in \mathbb{C}$ is **algebraic over K** if there is some nonzero polynomial $P \in K[x]$ such that $P(\alpha) = 0$.

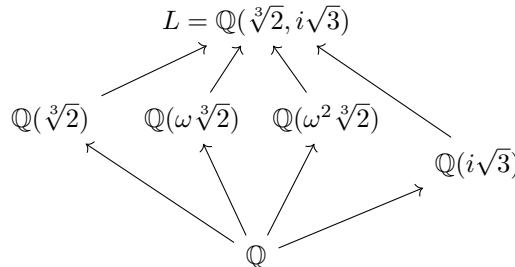
Consider $\mathbb{Q}(\sqrt{2})$, hours of highschool mathematics tells us we don't need to consider the entire large set of quotients, just:

$$\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}.$$

If we choose something **transcendental** (non-algebraic) like $\mathbb{Q}(\pi)$, then we must use the full quotient definition, and in this case we have $\mathbb{Q}(\pi) \cong \mathbb{Q}(X)$ the field of fractions of $\mathbb{Q}[X]$.

Definition 1.2.2. For some $f \in K[x]$ with distinct complex roots $a_1, \dots, a_n \in \mathbb{C}$, the **splitting field** of f is $L = K(\alpha_1, \dots, \alpha_n)$.

Let $K = \mathbb{Q}$ and $f = x^3 - 2$. The roots of f in $\mathbb{C}$ are $\sqrt[3]{2}$, $\omega\sqrt[3]{2}$, and $\omega^2\sqrt[3]{2}$, where ω is $(1 - i\sqrt{3})/2$. So the splitting field is $L = \mathbb{Q}(\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2})$ which can be simplified to just $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$. What are the intermediate fields between $\mathbb{Q}$ and L ?



Lots of fields you might guess, like $\mathbb{Q}(i\sqrt{3}\sqrt[3]{3})$ and $\mathbb{Q}(\sqrt[3]{3} + i\sqrt{3})$ happen to already be in this diagram. But we cannot yet prove this is everything. The length of the arrows is a clue, they relate to the dimension as $\mathbb{Q}$ -vector spaces and subgroups in the Galois correspondence.

³I know this as "Ferrari's method".

Theorem 1.2.3 (Fundamental theorem of Galois theory). The **Galois group** of a field extension $K \subset L$ is

$$G = \text{Gal}(L/K) = \left\{ \varphi : L \xrightarrow{\sim} L \mid \varphi|_K = \text{id}_K \right\}$$

and the eponymous Galois correspondence:

$$\{K \subset F \subset L\} \xleftrightarrow{\sim} \{H \leq G\}$$

$$F \longmapsto F^\dagger := \{g \in G \mid g|_F = \text{id}_F\} = G_F \cdot$$

$$H^* := \{\alpha \in L \mid H\alpha = \alpha\} \longleftarrow H$$

If one knows the Galois group is a supgroup of the permutation of all the roots, then for the case $L = \mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$, there is seemingly no way to distinguish the roots, so we expect $G = S_3$, which is luckily true.

Fields are complicated and hard, there are two operations that “cavort” via a weird distributivity law, and proving the classification of subfields of $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$ already feels pretty impossible. Galois theory allows us to move information from the easy theory of finite groups into the world of field extensions.

Proposition 1.2.4. G is a subgroup of $\mathfrak{S}_n$, where n is the degree of f .

Proof. We want to exhibit any $\sigma \in G$ as a permutation of the roots of f . Any such σ is already an automorphism, so we just need to check it sends roots to roots. If $f(\lambda) = 0$, for some $\lambda \in L$, then we can rearrange $f(\sigma(\lambda)) = \sigma(f(\lambda)) = 0$.

So we get a homomorphism $\rho : G \rightarrow \mathfrak{S}_n$. Any automorphism in the kernel must fix every root, and so all of L , thus is the identity. Therefore, ρ is injective and we get $G \leq \mathfrak{S}_n$. $\square$

I claim this will be enough Galois theory to prove that the above diagram for $x^3 - 2$ is everything! Because in this case, G is all of $\mathfrak{S}_3$. To prove this I just have to construct enough elements. In general, constructing elements of Galois groups is very hard. We’ve already accepted general fields are very baroque, general automorphisms between them will surely be even less manageable. There are some easy cases.

Proposition 1.2.5. For some a not a perfect square in K , the Galois group of the field extension $K \subset K(\sqrt{a})$ will be C_2 .

Proof. The automomorphism $a + b\sqrt{a} \mapsto a - b\sqrt{a}$ is certainly nontrivial, this is all of $\mathfrak{S}_2$. $\square$

Now consider the quadratic extension in the diagram above: $K = \mathbb{Q}(\sqrt[3]{2}) \subset L = K(i\sqrt{3})$, the generator of $\text{Gal}(L/K)$ is τ such that $\tau(i\sqrt{3}) = -i\sqrt{3}$, this is certainly part of all of G as if it fixes $\mathbb{Q}(\sqrt[3]{2})$ it definitely fixes $\mathbb{Q}$. If we now explicitly calculate how this permutes the roots of f , we see $\tau = (2\ 3)$.

Let’s now do the same with $\mathbb{Q}(\omega\sqrt[3]{2})$, we’ll see the generator of this Galois group is $(1\ 3)$. I don’t actually need to compute how this τ acts on the roots, certainly it fixes $\omega\sqrt[3]{2}$, and certainly it is nontrivial, so it can only be $(1\ 3)$.

These two roots generate all of $\mathfrak{S}_3$, so we have naively computed the galois group of $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$. BUT, we still haven’t show that there are no other fields missing from our diagram, we must still simply believe in the full Galois correspondence to continue.

2 Actual Stuff

All field homomorphisms are injective⁴ so I’ll write $\text{Emb}(K, L)$ ⁵ for $\text{Hom}(K, L)$ to remind us. Typically there is a fixed field k in the background, and all fields L come with a fixed embedding $i : k \hookrightarrow K$ which we normally suppress.⁶ If K, L are two such fields, we should only consider embeddings that fix k :

$$\text{Emb}_k(K, L) := \{f : K \rightarrow L \mid f|_k = \text{id}_k\}$$
⁷

In particular, K will be a k -vector space. We can gain a lot of information by thinking about these field extensions in term of representations over our Galois groups, we won’t talk about that here though.

⁴If $\phi : K \rightarrow L$ is a field homomorphism then $\ker(\phi) \trianglelefteq K$ is either 0, making ϕ injective, or K , a contradiction as $\phi(1) = 0$.

⁵Emb stands for embedding.

⁶We normally think of $k = \mathbb{Q}$, in which case there is only ever one embedding of $\mathbb{Q}$ into any algebraic extension.

⁷more specifically, f should commute with the two embeddings.

Definition 2.0.1. A field extension $K \subset L$ is **finite** if L is finite as a K -vector space. The **degree** $[L : K]$ of the extension is $\dim_K L$.

Proposition 2.0.2 (Tower Law). If $K \subset L \subset M$ is a tower of field extensions, then

$$[M : K] = [M : L][L : K].$$

Proof. TODO: finite case is just counting bases, this will suffice. □

2.1 Minimal polynomials

Definition 2.1.1. Given $K \subset L$, for all $a \in L$, the **minimal polynomial** of a in K is the monic generator of the kernel of the ring homomorphism:

$$\text{ev}_a : K[X] \rightarrow L \quad \phi(x) \mapsto \phi(a)$$

Somewhere in a previous algebra course, we have proved $K[X]$ is an Euclidean domain, so a PID.

The first isomorphism theorem for rings now tells us:

$$K[X]/(f) \cong \text{im}(\text{ev}_a) \subset L$$

so $\text{im}(\text{ev}_a)$ is an integral domain, and thus f is prime so irreducible, and so (f) is maximal and $\text{im}(\text{ev}_a)$ is a field. Thus $K[a]$, the K -algebra generated by a , is the same as $K(a)$ the field generated by a as defined earlier.⁸

Conversely, if $f \in K[X]$ is monic and irreducible, and L is the field $K[X]/(f)$, then $K \subset L$ is a finite field extension. If we set $a := [X] \in L$, then $L = K[a] = K(a)$, and $f(x)$ is the minimal polynomial of a . The degree of such an extension $[K : K(a)]$ is equal to the degree of the minimal polynomial.

Finally, for a field extension $K \subset L$ and all $a \in L$ we will always have:

$$\text{Emb}_K(K(a), L) = \{\text{roots of } f \text{ in } L\}$$

2.2 Splitting fields

The construction of a splitting field of a K -irreducible polynomial as $L = K[X]/(f)$ is still important. When $L \subset \mathbb{C}$ we can adjoin sufficiently many roots of f and have a nice constructive form, but we are still interested fields that are not part of $\mathbb{C}$.

For example, consider $\mathbb{F}_2$ and the irreducible polynomial $X^2 + X + 1 \in \mathbb{F}_2[X]$, we can form $L = \mathbb{F}_2[X]/(X^2 + X + 1)$ which we know has degree 2 of $\mathbb{F}_2$ so has 4 elements. Are there any other fields of 4 elements? If one exists it is certainly of characteristic 2 so contains $\mathbb{F}_2$, any remaining element not contained in $\mathbb{F}_2$ has a minimal polynomial of degree 2 which must be irreducible, this can only be $X^2 + X + 1$ so there is only one such field of size 4 up to isomorphism. We can call this $\mathbb{F}_4$.⁹ In general, we know the theory of finite fields to be somewhat well behaved and all formed in a similar fashion. What about infinite fields of finite characteristic? Consider

$$K := \mathbb{F}_2(t) = \text{Frac}(\mathbb{F}_2[t]) = \left\{ \frac{P(t)}{Q(t)} \mid P(t), Q(t) \in \mathbb{F}_2[t], Q(t) \neq 0 \right\}$$

and the polynomial $X^2 - t \in K[X]$, which is irreducible. We can certainly prove this is irreducible by naively constructing a root and finding a contradiction. Or we can endeavor to learn some irreducibility conditions for general polynomials.

Proposition 2.2.1 (Eisenstein criterion). If $f = X^n + a_1X^{n-1} + \dots + a_n \in \mathbb{Z}[X]$ and there exists some prime $p \in \mathbb{Z}$ such that $p \mid a_i$ and $p^2 \nmid a_n$, then f is irreducible.

Corollary 2.2.2. By Gauss' lemma we know this implies f is also irreducible over $\mathbb{Q}[X]$.

⁸This is somehow not very constructive, our proofs that $K[X]$ is a PID normally goes by the ACC, we have no algorithm for computing a^{-1} in $K[a]$.

⁹There is a certainly a Galois action that swaps the two roots of $X^2 + X + 1$, so while our field is defined only up to non-unique isomorphism. Is $\mathbb{C}$ the same? can we distinguish i from $-i$?

But the Eisenstein criterion is “totally rubbish” and in practice will rarely work.¹⁰ We could now use the Gauss lemma to simplify our naive strategy as we only have to search $\mathbb{F}_2[t][X]$, but in fact the analogue between $\mathbb{Z}$ and $K[X]$ holds for the Eisenstein criterion, and by using $p = t \in \mathbb{F}_2[t]$, we deduce $X^2 - t$ is irreducible. We can even observe $K[x]/(X^2 - t) \cong K$, which you should perceive as being very weird!

But now, in more generality:

Definition 2.2.3. Let K be a field and $f \in K[x]$ be irreducible. A **splitting field** of f over K is a field extension $K \subset L$ such that:

$$f(x) = \prod_{i=1}^n (X - \lambda_i) \in L[X]$$

splits completely in $L[X]$; and L is generated by $S = \{\lambda \in L \mid f(\lambda) = 0\}$ over K .

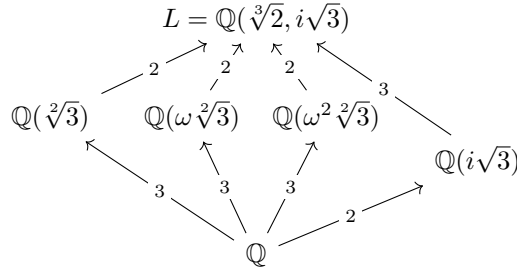
Splitting fields certainly exist, we can just progressively adjoin all roots of f until f splits completely. I claim, if:

$$\begin{array}{ccc} L_1 & & L_2 \\ & \cup & \cup \\ & K & \end{array}$$

are two splitting fields for f , then there exists some isomorphism $\phi \in \text{Emb}_K(L_1, L_2)$.

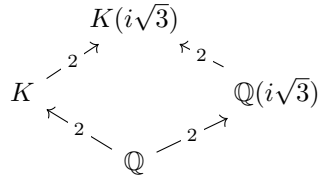
2.3 Motivation again

Back to the tower law to prove that we have a full subfield lattice:



TODO: I know the diagram is ugly, I don't care right now.

By the tower law, we now know $[L : \mathbb{Q}] = 6$, so any addition subfields K must have $[K : \mathbb{Q}] = 2$ or 3. Suppose $[K : \mathbb{Q}] = 2$, for K to be distinct it must not contain $i\sqrt{3}$. So $X^2 + 3$ remains irreducible over K and we can form the extension:



But the tower law now gives us $6 = [L : \mathbb{Q}] = [L : K(i\sqrt{3})][K(i\sqrt{3}) : \mathbb{Q}] = 4[L : K(i\sqrt{3})]$ and $4 \nmid 6$.

If instead $[K : \mathbb{Q}] = 3$, for K to be distinct it must contain no roots of $X^3 - 2$, so we can form $K[X]/(X^3 - 2)$, a degree three extension, and $9 \nmid 6$.

2.4 Back to splitting fields

Lemma 2.4.1. Let f be an irreducible polynomial in $k \subset K$, and $K(a) = K[X]/(f)$, then for any $K \subset L$:

$$\text{Emb}_K(K(a), L) \cong \{\text{roots of } f \text{ in } L\}$$

¹⁰It is like if you drop your keys on a dark street, the mathematician goes to look for your keys on the other side of the street where there is a lamp. Sure, *if* your keys happen to have reached the other end of the street, they will certainly be found. But this is unlikely, just how it is unlikely an irreducible polynomial satisfies Eisenstein.

Proof. consider the map which sends $\varphi \in \text{Emb}_k(K(a), L)$ to $\varphi(a) \in L$ we know φ will commute with f so $\varphi(a)$ is a root. Indeed, this map is injective as if we have some $\phi(a) = \phi'(a)$ in L , they must now agree on all of $K(a)$. If instead I have a root $b \in L$ of f , then the map $\text{ev}_b : K[X] \rightarrow L$ must have f the generator of the kernel, so the isomorphism theorem gives us a map $K(a) = K[X]/(f) \rightarrow L$. $\square$

We have already seen this to be true! There are three embeddings of $\mathbb{Q}(\sqrt[3]{2})$ into $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$ as either $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\omega\sqrt[3]{2})$, or $\mathbb{Q}(\omega^2\sqrt[3]{2})$.¹¹

Lemma 2.4.2. For K a field and $f \in K[X]$ and suppose there are two extensions $K \subset L_1$, which is generated by the roots of f , and $K \subset L_2$ where f splits completely, then

$$\text{Emb}_K(L_1, L_2) \neq \emptyset$$

Proof. If $K = L_1$, then we are done, otherwise choose some $a \in L_1 \setminus K$ s.t. $f(a) = 0$ and let $g \in K[X]$ be the minimal polynomial of a , certainly $g \mid f$, and so g splits completely in $L_2[X]$. There must be another root $b \in L_2$ such that $g(b) = 0$, this corresponds to an embedding $K(a) \subset L_2$ by the previous lemma. We now just need to show

$$\text{Emb}_{K(a)}(L_1, L_2) \neq \emptyset$$

but as $[L_1 : K(a)] < [L_1 : K]$, by induction we are done. $\square$

Corollary 2.4.3. For K a field and $f \in K[X]$ any two splitting fields of f are isomorphic over k .

2.5 Normal extensions

Definition 2.5.1. $k \subset K$ is **normal** if for all $k \subset \Omega$ and $x_1, x_2 \in \text{Emb}_k(K, \Omega)$ there exists a $\varphi \in \text{Aut}(K/k)$ such that $x_1 = x_2\varphi$

Intutively, the image of any k -embedding $K \subset \Omega$ is fully determined by the embedding $k \subset \Omega$. The shape of a normal extension around k is fully determined.

If, instead, we fix our ‘big’ field extensions $k \subset \Omega$, then $\text{Emb}_k(-, \Omega)$ becomes a functor from algebraic k -extensions to **FinSet**, and we always get a faithful **right** action of $\text{Gal}(K/k)$ on $\text{Emb}_k(K, \Omega)$ by pre-composition. Now $k \subset K$ is normal iff this action will be transitive, equivalently iff $\text{Emb}_k(K, \Omega)$ is a $\text{Gal}(K/k)$ -torsor.¹²

Proposition 2.5.2. If $f(x) \in k[X]$ and $k \subset K$ is the splitting field of f , then $k \subset K$ is normal.

Proof. Consider $K \subset \Omega$, we want to show for any $x : K \hookrightarrow \Omega$, $x(K) \subset K$. If $a \in K$ is a root of f , then $x(a)$ is also a root of f , so $x(a) \in K$, as K is generated by roots of f we must have $x(K) \subset K$. $\square$

The converse is true but we will prove this later. Splitting fields are fully determined by a polynomial, and somehow **all** of the interactions between the elements are controlled by this polynomial, so the ‘shape’ of K in any bigger extensions is already totally controlled by a split polynomial.

2.6 Separable extensions

Definition 2.6.1. $k \subset K$ is **separable** if for all towers of subfields

$$k \subset F_1 \subset F_2 \subset K$$

there exists a big field $k \subset \Omega$ with at least two distinct K_1 -embeddings $x, y : K_2 \rightarrow \Omega$. I’ll give a slightly down to earth criterion, but without proof for the moment.

If $k = \mathbb{F}_2(t) \subset K = k(\sqrt{t})$, the polynomial $X^2 - t \in k[X]$ is irreducible, and K is certainly its splitting field, but it only has one root in K . So, there is only one element of $\text{Emb}_{F_1}(F_2, \Omega)$. What is an example in the positive?

Definition 2.6.2. A polynomial of degree n is **separable** if it has n distinct roots in its splitting field.

¹¹“We rrrrrrip $\mathbb{Q}(\sqrt[3]{2})$ out of its environment, it is an animal embedded in the jungle of complex numbers”. It is just isomorphic to the abstract field $\mathbb{Q}[X]/(X^3 - 2)$, which exists as three different copies in $\mathbb{C}$ “This abstract copy can be incarnated as a lion or a tiger or a zebra”. We are just measuring the wasted space of the splitting field.

¹²Something that looks like $\text{Gal}(K/k)$ without a choice of identity.

TODO: polynomial derivative and jacobian criterion, fucking snooze

If a polynomial f is irreducible, when is it also separable, iff $D(f) = 0$, in characteristic 0 this never happens, but in characteristic $p > 0$ this is true iff $f(X) = h(X^p)$ for some $h \in k[X]$

Definition 2.6.3. A field F of characteristic p is perfect if all elements are perfect p th powers.

For such a field, the **Frobenius** map $\text{Fr}_p(a) = a^p$ is a field embedding. When $F = \mathbb{F}_p$, Fr is in fact the identity, as

$$a^p \equiv a \pmod{p}$$

Proposition 2.6.4. Finite fields are always perfect.

Proof. The Frobenius map is an injective endomorphism of a finite set, so is surjective and thus every element is realised as a perfect p th power. $\square$

Proposition 2.6.5. In a perfect field every polynomial is separable.

Proof. $f(X) = h(X^p) = g(X)^p$, as all the coefficients are perfect p th powers. $\square$

So we only discover irreducible and inseparable polynomials in infinite fields of characteristic $p > 0$.

2.7 Axioms

Proposition 2.7.1 (A1). For all $k \subset K \subset L$, the set $\text{Emb}_k(K, L)$ is finite, in fact $|\text{Emb}_k(K, L)| \leq [K : k]$.

Proposition 2.7.2 (A2). Every morphism is injective, and every endomorphism is also surjective.

Proposition 2.7.3 (A3). If $k \subset K_1, K_2 \subset L$, then $K_1 \cap K_2$ and $K_1 K_2$ (the smallest subfield containing K_1 and K_2) are both fields between k and L .¹³

Proposition 2.7.4 (A4). Suppose $k \subset K, L$ then there exists some finite extension $k \subset \Omega$ containing them both:

$$\begin{array}{ccc} & K & \\ & \swarrow \quad \searrow & \\ k & & \Omega \\ & \nwarrow \quad \nearrow & \\ & L & \end{array}$$

In general, Ω is not unique.

Proposition 2.7.5 (A5). If K is a field and G is a finite group acting faithfully on K as field automorphisms, then:

$$K^G = \{\alpha \in K \mid \forall g \in G, g(\alpha) = \alpha\}$$

is a field, and the map $G \hookrightarrow \text{Gal}(K/K^G)$ is an isomorphism.

TODO: phrase this properly and actually define the specific galois category we work in.

Proof of A4. We will go by induction on $[K : k]$ the bases case has $K = k$ so taking $\Omega = L$ works. Now suppose $[K : k] > 1$, then there exists some $\alpha \in K \setminus k$, this will have a minimal polynomial $f \in k[X]$, let L' be the splitting field of $f \in L[X]$, then $\text{Emb}_k(k(\alpha), L') = \{\text{roots of } f \text{ in } L'\}$ which must be nonempty as f is nontrivial. THus we can form the daigram:

$$\begin{array}{ccc} & & K \\ & & \swarrow \\ & k(\alpha) & \\ & \swarrow \quad \searrow & \\ k & & L' \\ & \nwarrow \quad \nearrow & \\ & L & \end{array}$$

where by the tower law $[K : k(\alpha)] < [K : k]$ so, by induction, we're done. $\square$

¹³The interesection is visibly the pullback, but the product isn't just the pushout, in the notes this is called a **framed pushout**, my understanding is that there is no actual pushout, only when we specify how K_1 and K_2 are embedded into a bigger field Ω there is a pushout in $k\mathbf{Gal} \downarrow \Omega$. What else do I name this category???

This one is very important

Lemma 2.7.6. Suppose $k \subset K$ is normal, and $k \subset F \subset K \subset \Omega$ is a tower, then the natural restriction:

$$\rho : \text{Emb}_k(K, \Omega) \rightarrow \text{Emb}_k(F, \Omega)$$

is surjective. Equivalently,

Proof. For any $x \in \text{Emb}_k(F, \Omega)$, we can form the diagram:

$$\begin{array}{ccc} K & \subset & \Omega' \\ \cup & \searrow & \cup \\ F & \xrightarrow{x} & \Omega \\ \cup & & \\ k & & \end{array}$$

where Ω' is produced by (A4), and as $k \subset K$ is normal, the inclusion $K \subset \Omega'$ factors through $K \subset \Omega$. $\square$

So when $k \subset K$ is normal, the left action $\text{Aut}(K/k) \curvearrowright \text{Emb}_k(F, K)$ by post-composition is transitive. Specifically, if $f \in k[X]$ is irreducible and splits completely in K , then $\text{Aut}(K/k)$ acts transitively on these roots, just by taking $F = k(a)$ for a some root in K .

Corollary 2.7.7. The natural composition with the given $K \subset \Omega$:

$$\text{Emb}_k(F, K) \rightarrow \text{Emb}_k(F, \Omega)$$

is also surjective.

Proof. By the previous lemma it suffices to consider $\sigma \in \text{Emb}_k(K, \Omega)$ restricted to F , but we already have a canonical $K \subset \Omega$, so σ is just represented by some element of $\text{Aut}(K/k)$ which we just restrict to F to produce our desired $\text{Emb}_k(F, K)$. $\square$

This says any k -embedding $x : F \rightarrow \Omega$ still lands F in K , this is a stronger statement of the definition of K being a normal extensions, in fact if we take $F = K$ we recover this.

A striking consequence of this corollary is that any irreducible $f \in k[X]$ either has no roots in K , or splits completely! Indeed if f has any root $a \in K$ then let Ω be the splitting field of f in K , we can form the tower $k \subset k(a) \subset K \subset \Omega$ and we have the isomorphism:

$$\{\text{roots of } f \text{ in } \Omega\} \cong \text{Emb}_k(k(a), \Omega) \cong \text{Emb}_k(k(a), K) \cong \{\text{roots of } f \text{ in } K\}$$